

Certificate of Analysis

[REDACTED]

Attention:
Report:
Client Reference:
Received Date:
Date Reported:

[REDACTED]
25 July 2024
19 August 2024

Lal Lal

METHODOLOGY:

Semi-volatile
Extraction
(SVOC Scan)

Aqueous samples were sequentially extracted with dichloromethane, concentrated under reduced pressure and temperature and then analysed by capillary gas chromatography with a mass spectrometer operated in full scan electron impact ionisation mode.

Solid samples were first extracted with acetone/dichloromethane and then subjected to the aforementioned analysis.

Mass Spectral
Library Search

The major peaks detected in the GC-MS chromatogram were analysed by Automated Mass Spectrometry Deconvolution and Identification System (AMDIS) using the NIST/EPA/NIH Mass Spectral Library (NIST 14) and NIST Mass Spectral Search Program (Version 2.2) (National Institute of Standards and Technology, U.S. Environmental Protection Agency/National Institutes of Health) library according to the guidelines described below. Library match is a percentage of the accuracy of the ion present in the sample against the NIST library record and is only a tentative identification. Definitive identification requires comparison with an authentic standard. The library contains ~275,000 mass spectra.

Environment Testing

After visual comparison of sample spectra with the nearest library searches a tentative identification were assigned based upon a quality match.

Guidelines for tentative identification are:

- (1) Relative intensities of major ions in the reference spectrum (ions > 10% of the most abundant ion) should be present in the sample spectrum.
- (2) The relative intensities of the major ions should agree within $\pm 20\%$. (Example: For an ion with an abundance of 50% in the standard spectrum, the corresponding sample ion abundance must be between 30 and 70%.)
- (3) Molecular ions present in the reference spectrum should be present in the sample spectrum.
- (4) Ions present in the sample spectrum but not in the reference spectrum were reviewed for possible background contamination or presence of co-eluting compounds.
- (5) Ions present in the reference spectrum but not in the sample spectrum were reviewed for possible subtraction from the sample spectrum because of background contamination or co-eluting peaks. Data system library reduction programs can sometimes create these discrepancies.

Environment Testing

Client Sample ID: TURBINE

Table 1: NIST Tentative Identification of the Top 9 Matches

Component RT	Compound Name	CAS#	Formula	Area	Match Score
1.8243	1,6-Octadiene, 5,7-dimethyl-, (R)-	85006-04-8	C10H18	354689	60.5
2.5571	Cyclopentasiloxane, decamethyl-	541-02-6	C10H30O5Si5	31432	89.6
3.0545	Cyclohexasiloxane, dodecamethyl-	540-97-6	C12H36O6Si6	47842	94.3
3.5626	3-Isopropoxy-1,1,1,7,7,7-hexamethyl-3,5,5-tris(trimethylsiloxy)tetrasiloxane	71579-69-6	C18H52O7Si7	20887	66.3
4.1349	Cyclooctasiloxane, hexadecamethyl-	556-68-3	C16H48O8Si8	12434	81.8
8.7401	Pentadecanal-	2765-11-9	C15H30O	52037	64.5
8.9327	Carbonic acid, heptadecyl prop-1-en-2-yl ester	1000382-90-2	C21H40O3	18847	58.9
9.5905	Heptacosane	593-49-7	C27H56	42368	55.7
10.4677	5-Cholesten-3.beta.-ol-7-one, methyl ether	1000332-98-5	C28H46O2	113362	57.4

Environment Testing

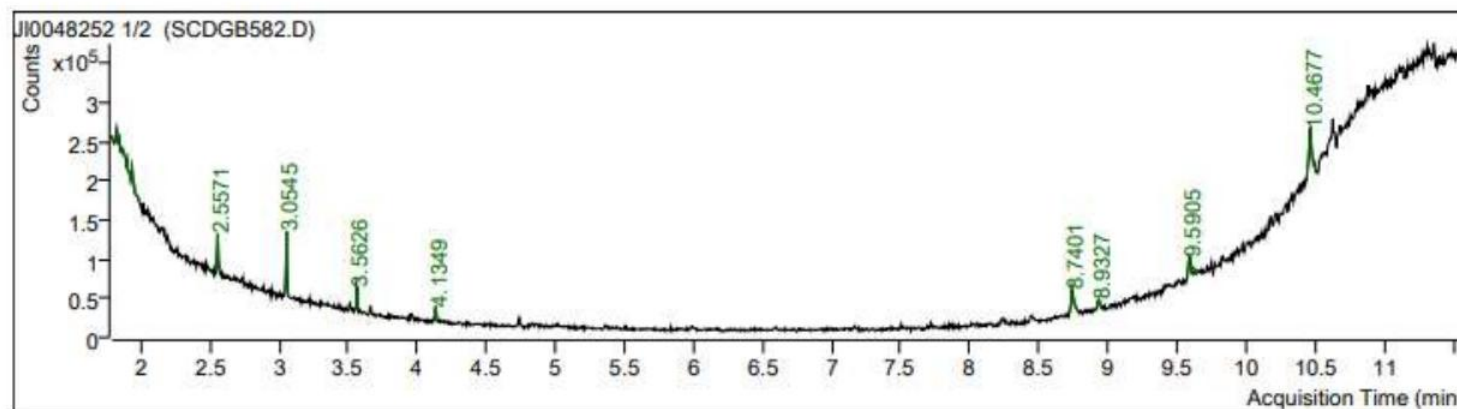


Figure 1: Total Ion Chromatogram (TIC) for [REDACTED] Environment Testing Sample No: [REDACTED]

NOTE: Numbered peaks in this TIC refer to those compounds identified in Table 1: NIST Tentative Identification of the Top 9 Matches and do not include internal or surrogate standards.



Environment Testing

Sample History

Where samples are submitted/analysed over several days, the last date of extraction and analysis is reported.

A recent review of our LIMS has resulted in the correction or clarification of some method identifications. Due to this, some of the method reference information on reports has changed. However, no substantive change has been made to our laboratory methods, and as such there is no change in the validity of current or previous results (regarding both quality and NATA accreditation).

If the date and time of sampling are not provided, the Laboratory will not be responsible for compromised results should testing be performed outside the recommended holding time.

Description		Testing Site	Extracted	Holding Time
SVOC	US EPA 8270	Melbourne	27 July 2020	14 Days

Environment Testing Internal Quality Control Review and Glossary

General

1. Laboratory QC results for Method Blanks, Duplicates, Matrix Spikes, and Laboratory Control Samples are included in this QC report where applicable. Additional QC data may be available on request.
2. All soil results are reported on a dry basis, unless otherwise stated.
3. Actual PQLs are matrix dependant. Quoted PQLs may be raised where sample extracts are diluted due to interferences.
4. Results are uncorrected for matrix spikes or surrogate recoveries.
5. SVOC analysis on waters is performed on homogenised, unfiltered samples, unless noted otherwise.
6. Samples were analysed on an 'as received' basis.
7. This report replaces any interim results previously issued.

Holding Times

Please refer to 'Sample Preservation and Container Guide' for holding times (QS3001).

For samples received on the last day of holding time, notification of testing requirements should have been received at least 6 hours prior to sample receipt deadlines as stated on the Sample Receipt Advice.

If the Laboratory did not receive the information in the required timeframe, and regardless of any other integrity issues, suitably qualified results may still be reported.

Holding times apply from the date of sampling, therefore compliance to these may be outside the laboratory's control.

****NOTE:** pH duplicates are reported as a range NOT as RPD

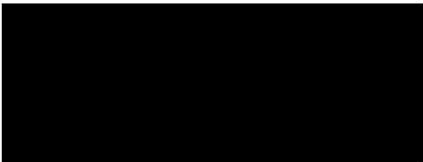
UNITS

mg/kg: milligrams per kilogram
µg/L: micrograms per litre
ppb: Parts per billion

mg/L: milligrams per litre
ppm: Parts per million
%: Percentage

TERMS

Dry	Where moisture has been determined on a solid sample the result is expressed on a dry basis.
LOR	Limit of Reporting.
SPIKE	Addition of the analyte to the sample and reported as percentage recovery.
RPD	Relative Percent Difference between two Duplicate pieces of analysis.
LCS	Laboratory Control Sample - reported as percent recovery
CRM	Certified Reference Material - reported as percent recovery
Method Blank	In the case of solid samples these are performed on laboratory certified clean sands. In the case of water samples these are performed on de-ionised water.
Surr - Surrogate	The addition of a like compound to the analyte target and reported as percentage recovery.
Duplicate	A second piece of analysis from the same sample and reported in the same units as the result to show comparison.
Batch Duplicate	A second piece of analysis from a sample outside of the clients batch of samples but run within the laboratory batch of analysis.
Batch SPIKE	Spike recovery reported on a sample from outside of the client's batch of samples but run within the laboratory batch of analysis.
USEPA	United States Environmental Protection Agency
APHA	American Public Health Association
ASLP	Australian Standard Leaching Procedure (AS4439.3)
TCLP	Toxicity Characteristic Leaching Procedure
COC	Chain of Custody
SRA	Sample Receipt Advice
CP	Client Parent - QC was performed on samples pertaining to this report
NCP	Non-Client Parent - QC performed on samples not pertaining to this report, QC is representative of the sequence or batch that client samples were analysed within
TEQ	Toxic Equivalency Quotient



Environment Testing

Comments

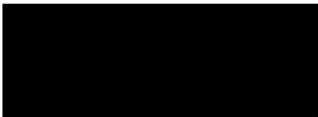
Sample Integrity

Custody Seals Intact (if used)	N/A
Attempt to Chill was evident	Yes
Sample correctly preserved	Yes
Organic samples had Teflon liners	Yes
Sample containers for volatile analysis received with minimal headspace	Yes
Samples received within Holding Time	Yes
Some samples have been subcontracted	No

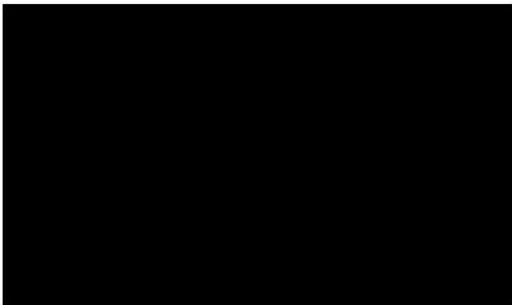
Qualifier Codes/Comments

Code	Description
N/A	Not applicable

Authorised by



Analytical Service Manager
Senior Analyst



General Manager

Final Report – this report replaces any previously issued Report.

